ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/337981927

# 5G V2X Architecture and Radio Aspects

**Conference Paper** · October 2019

DOI: 10.1109/CSCN.2019.8931319

CITATIONS
5

READS
2,901

**5 authors**, including:

Dimitrios Karampatsis
Lenovo Technology UK
**4** PUBLICATIONS **15** CITATIONS

SEE PROFILE

Andreas Kunz
Lenovo
**41** PUBLICATIONS **672** CITATIONS

SEE PROFILE

**Some of the authors of this publication are also working on these related projects:**

Project 3GPP SA3 View project

Project Special Issue on 5G Enhancements View project

All content following this page was uploaded by Andreas Kunz on 15 January 2020.

The user has requested enhancement of the downloaded file.

# 5G V2X Architecture and Radio Aspects

Karthikeyan Ganesan[1], Prateek Basu Mallick[1], Joachim Löhr[1], Dimitrios Karampatsis[2], Andreas Kunz[1]

Lenovo, Oberursel, Germany[1]
Lenovo, Basingstoke, United Kingdom[2]

email: {kganesan, pmallick, jlohr, dkarampatsis, akunz}@lenovo.com

*Abstract —*

3GPP standardization is evolving beyond 5G the New Radio air interface (NR-Uu) framework defined in 3GPP Release 15 (also known as 5G phase 1) to study the support of advanced vehicle to everything (V2X) services. The NR Sidelink (SL) being designed in 3GPP Release 16 (5G phase 2) to support advanced V2X services directly between vehicles and other road users is expected to finish by the end of 2019. The advanced V2X services primarily focuses on platooning, advanced driving, extended sensor and remote driving use-cases. Supporting an end-to-end latency of 3ms and high reliability up-to 99.999% requirement requires substantial improvements in the air-interface design. Long Term Evolution (4G LTE) V2X sidelink supports the delivery of Basic Safety Messages to nearby V2X User Equipments (UE, the mobile device), hence the sidelink physical layer and the higher protocol layers are designed to support only broadcast type traffic without Hybrid Automatic Repeat Request (HARQ) feedback. With the advanced V2X service requirement, NR sidelink design should study V2X message delivery for different cast types such as unicast, groupcast and broadcast type traffics that requires substantial changes starting in V2X Release 16 network architecture, physical and protocol layers. This paper provides insights on the current Release 16 NR sidelink design for V2X. The network architecture enhancement discusses supporting flow- based Quality of Service (QoS) model for NR sidelink and the newly introduced network entities such as Network Data Analytics Function (NWDAF) providing QoS analytic. The design of the NR sidelink physical layer based on the Cyclic-Prefix Orthogonal Frequency-Division Multiplexing (CP-OFDM) waveform addresses the necessary improvements needed in the sidelink procedures for reliable packet transmissions in a certain communication range. The protocol layer introduces a new overlapping zone concept to study both about HARQ feedback in a communication range and the resource segregation.

***Index Terms – V2X, 3GPP, 5GS, NR SL***

## I. INTRODUCTION

The automotive industry is in the midst of a transition towards automated driving and the development of advanced driver-assistance systems (ADAS) where vehicles can quickly react to complex changing driving environments is stimulating the move to a digital cooperative intelligent transportation system (ITS). Cooperative ITS, a transportation system in which all road users, including pedestrians cooperate with each other, promises to increase the traffic efficiency, road safety and reduce the number of road traffic accidents. In order to achieve this ambitious goal, vehicles should have information of their surroundings, process information to e.g. predict some potential hazard and take protective actions accordingly. The present sensor based technology such as Radar, Lidar, and camera limits the local and individual perception of the vehicle and hence cannot work in non-line-of-sight (NLOS), bad weather etc., [1]. However, V2X as a complementary technology improves the vision beyond the range of sensor providing a 360-degree environmental awareness around the vehicle by reliably exchanging messages with other vehicles, road users and roadside units. V2X will play a pivotal role in enabling the deployment of fully autonomous vehicles with its ability to gather data from further ahead, e.g. hidden threats, to deliver a more predictable driving experience by expanding the scope of the vehicle's perception.

In this respect, a mobile communication system is required to ensure reliable exchange of road traffic related data even at high speeds. C-V2X (Cellular Vehicle-to-Everything) is a term introduced for cellular technologies optimized for transportation and connected vehicles. In particular, the cellular refers to both 4G LTE and 5G NR (New Radio) releases of specifications, whereas the "everything" refers to the variety of entities/users vehicles such as Vehicle-to-Vehicle (V2V), Vehicle-to-Pedestrian (V2P), Vehicle-to-Infrastructure (V2I), Vehicle-to-Network (V2N).

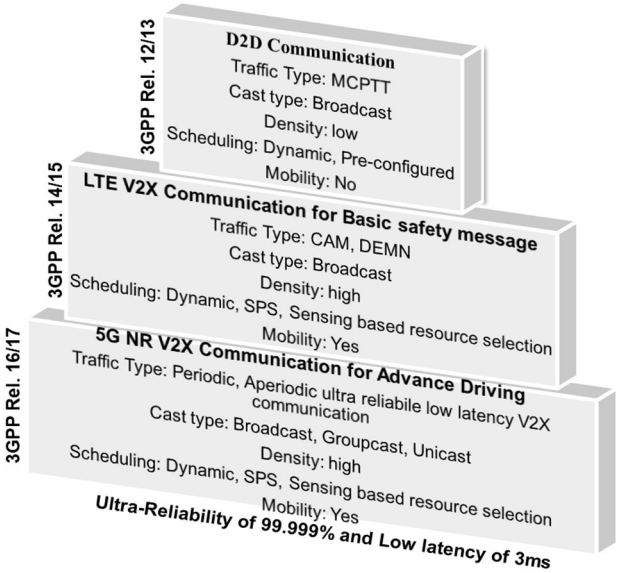


***Figure 1: 3GPP Sidelink Evolution***

Initially, 3GPP Long Term Evolution (LTE) specified device-to-device (D2D) communications as part of (Proximity Services) ProSe services in Release 12 and Release 13. The PC5 interface, also referred to as the sidelink (SL) provided physical layer support mainly for D2D communication. In Release 14/15 as part of introducing LTE-V2X (Vehicle–to-everything) communication service, sidelink design enhanced specifically addressing high speed and high-density scenarios as shown in *Figure* 1. This feature sets the starting point for the evolution of vehicular applications not previously supported by mobile communication technology, paves the way for future-proof connectivity in the automotive domain [2].

LTE V2X supports the delivery of basic safety messages (BSM) like cooperative awareness message (CAM) and decentralized environmental notification message (DENM). NR V2X as part of Release 16 is supposed to support a range of challenging high-end V2X services, including very precise positioning and ranging to enable cooperative and automated driving that requires ultra-reliability and low latency. NR V2X is destined as C-V2X phase 3 supporting advanced V2X services apart from LTE Release 15 V2X. NR-V2X will complement and co-exist with LTE-V2X. 3GPP has defined the requirements for support of enhanced V2X use cases in NR V2X, which has four use case groups [3]. Vehicle platooning, Advanced driving, Extended sensors, Remote driving. The advanced V2X services envisioned for NR V2X (Release 16) require an enhanced design of the NR sidelink (NR-PC5) to meet the stringent reliability and latency requirements mentioned in *Table 1*.

This paper describes potential technical enhancements relevant in Release 16 for the NR sidelink to meet the requirements of the enhanced V2X services. The effort of facilitating an ultra-reliable 5G network with low latency capabilities is critical for realizing V2X communication services, which are paramount for road safety.

The structure of the paper is as follows, in Section II, we introduce the background of V2X in 5G from the system architecture point of view. Then Section III describes the 5G NR V2X sidelink enhancements for high reliability and latency. Section IV highlights the upcoming challenges for security aspects. Finally, we conclude in Section V providing also a brief discussion on the outlook.

| **V2X use-case description** | **Min End to end Latency (msec)** | **Max Reliability (%)** | **Minimum communication range (meters)** |
|---|---|---|---|
| **Platooning:** Cooperative driving with information exchange between platoon leader and group of vehicles | 10 | 99.99 | 80-350 |
| **Advanced Driving:** Vehicle shares its driving intention with vehicles in proximity such as cooperative maneuvers and Emergency trajectory alignment between vehicles | 3 | 99.999 | 360-700 |
| **Extended Sensor:** Exchange of raw or processed data gathered through local sensors or live video images among vehicles | 3 | 99.999 | 50-1000 |
| **Remote Driving:** Remote driver or a V2X application to operate a remote vehicle for those passengers who cannot drive by themselves or remote vehicles located in dangerous environments | 5 | 99.999 | N.A. |

***Table 1: V2X Use-case description and KPIs***

## II. 5G V2X NETWORK ARCHITECTURE OVERVIEW

3GPP Release 14 addresses V2X communication within the EPS architecture offering advantages compared to dedicated short-range communication (DSRC). Support for V2X services via utilising mobile network infrastructure and via direct communication between devices is possible.

In Release 16, 3GPP is defining enhancements for the V2X architecture, incorporating the V2X services within the 5G architecture as shown in *Figure* 2 (in red the main elements supporting V2X) taken from [4] [5]. The main functions and reference points supporting V2X are the following:

- *PC5 reference point:* The reference point between UEs for V2X communication. V2X communication over PC5 support LTE and/or NR radio.
- Uu reference point: The reference point between the UEs and the NG-RAN node. V2X communication over Uu support LTE and/or NR radio.
- *PCF (Policy Control Function):* Providing V2X configuration (radio parameters, QoS, Public Land Mobile Network support configuration) for V2X communication over PC5/Uu to V2X devices over control plane signalling (NAS signalling)
- NWDAF (Network Data Analytics Function): Provide QoS analytics information to enhance V2X communications over Uu

Through the 5G architecture (and its services), the following enhancements were possible for V2X:

- Support of enhanced V2X communication over Uu, leveraging the support of NR radio, Mobile Edge Computing and Ultra Reliable Low Latency communications (URLLC) allowing for low latency communications (e.g. support of Edge Computing for autonomous driving).
- Support of enhanced V2X communication over PC5 over multiple 3GPP Radio Access Technologies (RATs, i.e. LTE and NR) allowing for better QoS support over NR radio.

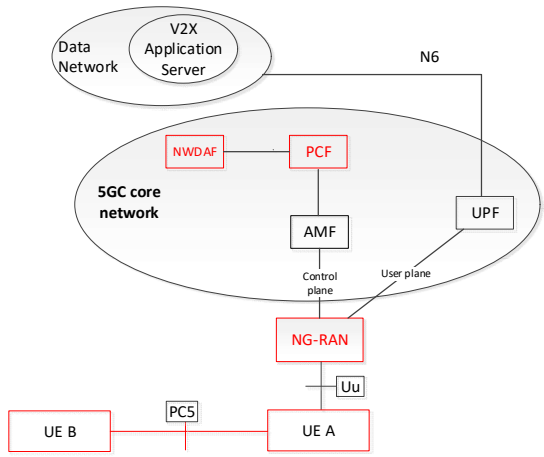


***Figure 2: Release 16 V2X architecture within 5G system architecture***

### *A. QoS support for V2X communication over PC5/Uu*

For V2X Communication over Uu, the network data analytics function defined for the 5G architecture provides QoS degradation awareness [6]. A V2X application server may request from the 3GPP network information about potential QoS change for an indicated geographic area and time interval in order to adjust the application behaviour in advance. The NWDAF function within the 3GPP network collect measurement data and derive statistical information about potential QoS change that are provided in advance to the V2X application server.

For V2X communication over PC5 over NR a major change to earlier releases is the QoS model. In Release 14 V2X, QoS supports Release 13 ProSe QoS mechanism by using the ProSe per packet priority (PPPP). The PPPP is a scalar value (range of 8 possible values) that defines the transmission priority of a V2X packet for the sidelink. Either the V2X application or pre-configuration (based on the V2X configuration information) provides PPPP for the UE.

In Release 15 V2X, the PPPP mechanism was further enhanced by introducing the ProSe per packet reliability (PPPR) that allows Access Stratum (AS) layer in the UE to identify if a V2X packet needs to be reliably transmitted (i.e. duplicated) over sidelink by multiple carrier. V2X service type to carrier mapping information is provided to the AS layer and AS layer transmit the V2X packet on a given carrier based on this information.

In Release 16, 3GPP adopted the same QoS model used over Uu (i.e. flow-based QoS) in order to support the enhanced QoS supported with NR sidelink as shown in *Figure* 3. With the QoS-flow based mechanism, for each V2X packet transmitted over sidelink contains a QoS flow is associated with a specific QoS parameters. The UE determines how to associate a V2X packet to a QoS Flow based on PC5 QoS rules provided by the network (i.e. PCF).

The QoS parameters defined for sidelink/PC5 communication are:

- A set of standardised QoS characteristics indicated through the PQI value [7]. Each PQI value is associated with specific resource type (GBR, delay critical GBR or non-GBR), priority level, packet delay budget, packet error rate, averaging window (for GBR only) and Maximum Data Burst Volume (for GBR only).
- PC5 Flow Bit Rates: Guaranteed or Best effort flow bit rates
- PC5 Link Aggregate Bit Rates: Defines the maximum aggregate bit rate over all unicast links
- Range parameter: Defines the minimum distance that the QoS parameters are applicable.

An illustration of the Release 16 QoS model over sidelink is shown in *Figure* 3 where the Service Data Adaptation Protocol (SDAP) layer in the AS maps the QoS flows to corresponding radio bearer.

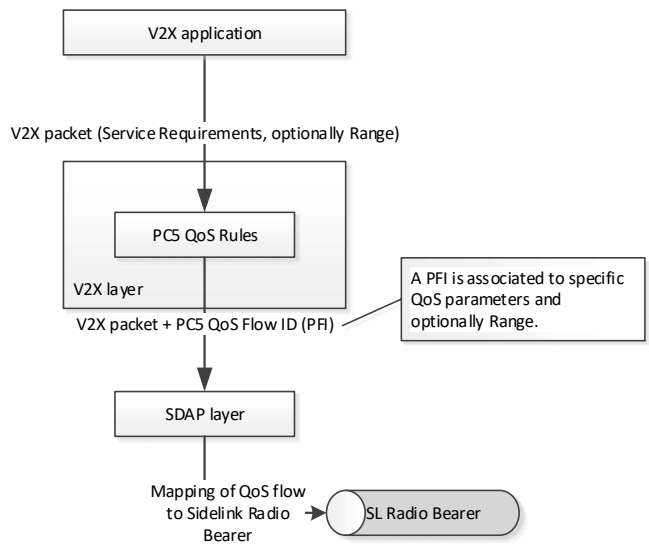


***Figure 3: Flow-based QoS model over PC5***

## III. Radio Aspects Enhancements for 5G V2X

The NR sidelink physical layer and protocol layer design requires enhancements to support different cast types (unicast, groupcast and broadcast), ultra-reliability and lower latency to transmit both periodic and aperiodic traffics. The NR sidelink design supports Cyclic-Prefix Orthogonal Frequency-Division Multiplexing (CP-OFDM) however the LTE V2X sidelink, uses Single Carrier Frequency Division Multiple Access (SC-FDMA). The NR Rel-15 framework is scalable to an extreme variation of requirements by using different subcarrier spacing and a flexible slot-based frame structure. It addresses for example diverse spectrum, deployments and services like eMBB (enhanced Mobile Broadband) or URLLC (Ultra reliable low latency) services. The NR Release 16 sidelink design focuses mainly on the reliability improvement and reduced processing latency.

Similar to Release 14, the V2X UEs support two modes of operation for V2X communication over the sidelink: NR mode 1 where RAN schedules sidelink resources and NR mode 2 where sidelink UE autonomously selects resources from a pre-configured or RAN signalled resource set:

In NR mode-1 operation, the RAN node centrally allocates resources for V2X communication over sidelink. Such mode requires the UE to interface with the RAN node every time the UE initiates a V2X communication over sidelink, i.e. the V2X UE requires being in network coverage.

In NR mode-2 operation, the V2X UE selects the resources for V2X communication autonomously. V2X UE perform sensing on a set of resources, i.e. resource pool, to avoid collision for V2X message transmission. Geographical zone based segregation of resource pools avoids collision during message transmission. This mode of operation enables V2X UE(s) to communicate outside network coverage.

### *A. NR Sidelink Physical layer Overview*

The NR sidelink design supports different numerologies like 15KHz, 30KHz and 60KHz Subcarrier Spacing (SCS) for frequency range 1 (FR1) which is up-to 6 GHz while Subcarrier Spacing of 60KHz, 120KHz are supported for FR2 (6 GHz to 52.6 GHz) and normal cyclic prefix (CP) whereas Extended CP support is added only for 60KHz SCS. The support of different numerology in the system ensures that the higher subcarrier spacing for V2X provides robustness against Doppler whereas extended CP provides robustness against high delay spread as shown in [8] .

#### *A.1. NR Sidelink Physical layer structures*

In NR sidelink design, Physical Sidelink Control Channel (PSCCH) carries the sidelink control information (SCI) and the associated Physical Sidelink Shared Channel (PSSCH) carrying the sidelink data are time domain multiplexed. The part of PSCCH and the associated PSSCH are transmitted in adjacent symbol as shown in *Figure* 4 unlike in LTE sidelink uses frequency domain multiplexing. Receiver UE(s) utilize the first symbol in the slot allocated for automatic gain control (AGC) as part of pre-processing to normalize the incoming signal power. The high reliability of NR sidelink unicast and groupcast transmission requires a new Physical Sidelink Feedback Channel (PSFCH) to carry sidelink HARQ feedback in the last OFDM symbol of a slot. In order to perform efficient link adaptation, feedback of the channel quality information (CQI) is required from the Receiver (RX) UE. NR sidelink retains the concept of sub-channel and resource pool with the flexibility of configuring resource pools with or without PSFCH resources [9]. NR V2X supports two stage SCI as part of enabling forward compatibility with future V2X release and simplifying the sensing operation of the V2X UE. The main advantage is that the V2X UE is not required to decode multiple SCI formats (i.e. SCI format for each cast type) during sensing operation. The design of two stage SCI enables splitting the SCI payload thereby transmitting the sensing related payload information like QoS priority of the data, occupied resource blocks, resource reservation interval etc., in the first stage of the SCI and as a broadcast transmission. The second stage of the SCI transmission carries information related data decoding of Physical Sidelink Shared Channel (PSSCH).

#### *A.2. NR Sidelink Physical layer procedures*

SA2 specifies minimum communication range (MCR) along with QoS as a Transmitter-Receiver (TX-RX) distance to maintain a given reliability of a groupcast packet transmitted from higher layer. Receiver UE(s) within the minimum communication range from the transmitter UE transmit SL HARQ feedback in PSFCH in response to the decoded transport block (TB). Following two options for groupcast feedback mechanism are studied. In the first option (Option-1), receiver UE(s) transmits HARQ-Negative Acknowledgement (NACK) on a common feedback resource whenever it fails to decode a TB. In the second option (Option-2), receiver UE(s) either transmit HARQ-Acknowledgement (ACK) if it successfully decode a TB or HARQ-NACK if it fails to decode a TB, on a given set of dedicated feedback resources allocated for each RX UE(s) in a group. The TX UE broadcasts in SCI both the MCR parameter and its present location in the form of a Zone id and this information is used by RX UE(s) to calculate the TX-RX distance, to determine if the transmission of HARQ feedback is required accordingly when the Tx-Rx distance is less than the MCR.

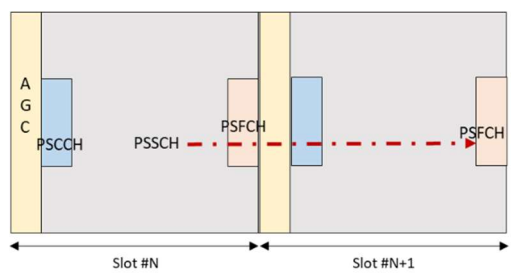


***Figure 4: Showing NR sidelink physical layer structure***

The TX UE decides the feedback option based on e.g. the reliability and latency requirement of the V2X message, number of RX UE(s) in a group, the amount of available feedback resources etc. In case of groupcast feedback option 2, the TX UE could identify RX UE(s) if the feedback resources are UE specific way and therefore it could try to enhance reliability by retransmitting the same TB in a unicast transmission to RX UE(s) with beamforming to the Rx UE that sent NACK feedback. In NR mode 1, as shown in the *Figure* 5, downlink control information (DCI) transmitted by the gNB to the TX UE for allocating sidelink resources also indicates the feedback timing and the resource required by the TX UE to transmit an indication to gNB to inform the need for the re-transmission resource of a TB.

### *B. NR Sidelink Protocol Overview*

In addition to the physical layer enhancements, a number of protocol layer enhancements are required to meet the kind of reliability needed for advance V2X applications. One straightforward way would be to use a high number of blind

HARQ re-transmissions made by the transmitter for a V2X message requiring high reliability.

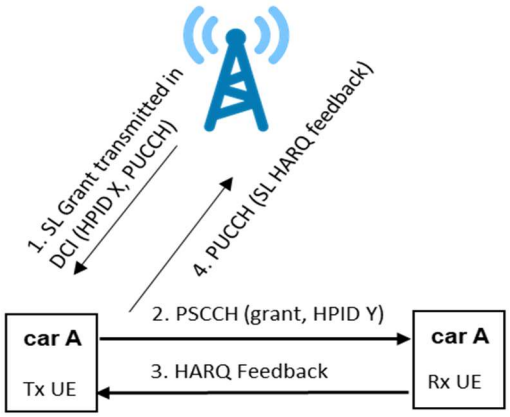


***Figure 5: Downlink control information transmitted from gNB indicates resource for transmitting sidelink HARQ feedback***

This would be wasteful by unnecessary transmission of same TB and the incurred inefficiency can be too high from a System perspective. To increase radio efficiency while not sacrificing the reliability (and within the latency constraint), some possible enhancements such as early stop indication of transmitting ACK to the gNB about the SL HARQ feedback status are currently under discussion.

First, a concept of Overlapping Zone configurations enables to use the usage of zone to both enable HARQ feedback mechanism and then resource allocation. A zone based Resource Pool allocation concept introduced as part of LTE V2X mode 4 operation requires enhancements to support NR groupcast feedback based on TX-RX distance. Zones can be used for the TX-RX distance based HARQ feedback purpose, where for example, the transmitter announces its Zone id (e.g. in SCI) and the receivers having determined their own Zone id calculate the distance from the transmitter determines about transmitting the HARQ feedback. For example, the RX UE transmits the HARQ feedback only if the distance is less than the signalled MCR, otherwise not. For this purpose, the Zone size should be small enough to minimize the "quantization loss" in the distance calculation. However, a smaller Zone would lead to frequent resource selection by a moving TX UE. In some cases, the transmitter may not even have enough time to finish sensing before it moves on to the next Zone; or, the remaining useful time might be minimal. This frequent Zone changes can lead to unacceptable V2X performance degradations due to frequent interruption.

In order to avoid those negative effects, NR V2X could study further about the overlapping zones where the first Zone configuration can typically smaller in size e.g. of 5m (or even smaller) width and length to minimize the distance inaccuracy between TX and RX UE(s) for the HARQ feedback. The second Zone configuration are bigger in size e.g. of 100 or 200 m (or even larger) width and length to allocate resources for V2X communication to UEs to avoid frequent resource pool changes as shown in the *Figure* 6. Second, the HARQ procedure can benefit from a number of enhancements for the Uu interface (Mode 1) and for the sidelink/PC5 interface (for both Mode 1 and Mode 2). In both cases, transmission errors on the air interface (leading to an erroneous HARQ feedback detection at the receiver) can be protected against by for example using an analysis of what HARQ feedback the UE transmitted last and what NDI (toggled/ not-toggled) has been received in the DCI with respect to the current NDI at the UE.

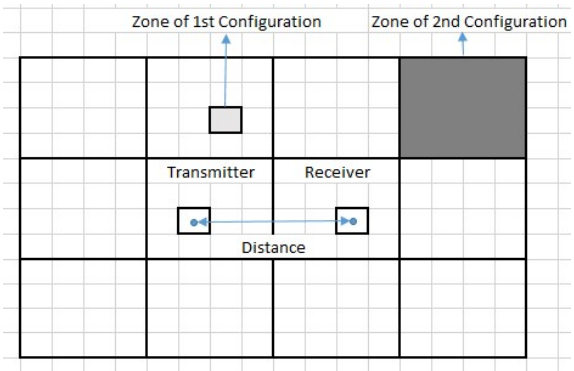


***Figure 6: Overlapping Zone concept***

Third, a parallel operation of Mode 1 and Mode 2 should allow the UE to switch between the modes when resources for transmission and/ or re-transmissions are not available in one of the Modes.

Further, the HARQ feedback assumes the knowledge of feedback resources at the receiver side, it is imperative to understand how the receiver UE(s) attain the HARQ feedback resources. The transmitter UE can distribute the Option-1 resources in a groupcast manner. Option-2 resources configures in a UE specific manner to the individual receiver UEs. This is especially important if the transmitter is interested to understand if some receiver UE may even have failed to receive the PSCCH (SCI) and therefore they could not transmit any feedback (DTX). For such UE specific feedback distribution, a PC5 RRC Connection between the transmitter and (towards) each of the receiver UE(s) needs to be assumed. However, establishing and maintaining a PC5 RRC Connection context can be challenging in PC5 framework since there is no easy means to unambiguously identify any particular UE as the V2X layers identities can keep changing and cannot identify the Physical UE.

## IV. Security Aspects

As described in the previous chapters, the NR based PC5 reference point is enhanced for V2X communication for groupcast and unicast mode, in addition to the connectionless broadcast on Access Stratum (AS) layer already available for EUTRA based PC5 V2X communications. Those new modes of communication have new security impacts that are currently studied in the 3GPP SA3 working group (security group) in [10]. For example the unicast mode over PC5 involves control plane signaling over two layers, the V2X layer and AS layer, the groupcast is performed by the application layer and UE configuration provisioning for V2X communication may be all subject for security protection and privacy considerations. For this reason, several key issues were identified seeking solutions. At the time of writing this

paper, only the key issues with their potential security threats were documented in the technical report [10].

### *A. Privacy protection for unicast messages over PC5*

The baseline solution for normative work in the system architecture group [7], proposes that upon an upper layer trigger or a timer, the UE updates in a Link Identifier Update Request message its peer with its new identifier(s) (Layer 2 Identifiers, L2 IDs), which gets acknowledged by the peer so that both UEs can use the new identifier(s). An attacker could be able to bind L2 IDs with long term V2X IDs and would result in privacy issues, i.e. trackability and linkability attacks.

### *B. Security for eV2X unicast messages over PC5.*

For 5G V2X communication, a new V2X service oriented L2 link establishment procedure was introduced where an initiating UE sends a Direct Communication Request message via broadcast to all UEs, and those interested in the V2X service reply to the initiator UE to establish a unicast communication. The underlying security should be based on the Proximity Services (ProSe) one-to-one link establishment, defined in [11]. If an attacker is able to perform a man-in-the-middle attack by breaking into the signalling exchange during the link establishment, then he could eavesdrop on signalling, data traffic, break the integrity etc.

### *C. Privacy protection for multicast messages over PC5*

Groupcast is called in the SA3 WG multicast and refers to the same procedures. The problem here is to prevent L2 ID tracking issues resulting form long duration multicast sessions.

This key issue is handling the same linkability and trackability problems a described in clause A but here for groupcast instead of unicast.

### *D. Security of identifier conversion in group communication*

A conversion procedure in the UE to determine the destination L2 ID can be performed based on mapping or configuration, in case it is not provided by the V2X application. This conversion should be secure that privacy and traceability aspects are fulfilled, else an attacker could link to a UE group memberships or the presence of group members in a certain location.

### *E. Security for setting up multicast*

At the moment there are no considerations for the security aspects of the multicast (groupcast) communication setup. An attacker may perform a man-in-the-middle attack on the L2 signalling in such a way that UEs receive wrong multicast information or no multicast information at all.

### *F. Security of the UE service authorization and revocation*

Service authorization and revocation should be secured over the PC5 reference point, in order to avoid that compromising the authorization by an attacker would leads to compromising the overall security of the service. It is still under discussion whether revocation is part of the study.

### *G. Cross-RAT PC5 control authorization indication*

With introduction of V2X for 5G there is for the cross-RAT PC5 control authorization no explicit or implicit indication defined on how the control from the cellular network for the LTE and/or NR sidelink is performed (via LTE Uu or NR Uu). Without an indication, the service authorization to NG-RAN might become problematic.

## V. Conclusions and Outlook

This paper provides an overview on V2X standardization on NR FR1 sidelink design currently studied as part of 3GPP NR Release 16. NR V2X improves Network architecture, Security, Physical layer and Protocol aspects considering the reliability and low latency requirement. Improvement in the network architecture are briefly discussed considering flow based QoS and QoS analytic from the NWDAF entity. The reliability improvement on the physical layer focusing on different HARQ feedback options, TX-RX distance method and resource allocation based on overlapping zone concept are detailed. Further improvement for NR V2X in Release 17 is set to study about the FR2 physical layer design, beamforming and Multiple Input Multiple Output (MIMO) with higher rank based on the distributed multiple antenna panels mounted on the vehicles. Various relaying aspects such as UE to network relaying and UE to UE relaying are under consideration for coverage extension to support some of the advanced V2X services in Release 17.

## References


[1] 5G Automotive Association, "The case for cellular V2X for safety and cooperative driving", White Paper, November 2016 URL: http://5gaa.org/pdfs/5GAA-whitepaper-23-Nov-2016.pdf

[2] S. Husain, A. Kunz, A. Prasad, K. Samdanis, J. Song, "Overview of 3GPP Standards for Vehicle-to-Everything", IEEE Conference on Standards for Communications and Networking (CSCN), September 2017.

[3] 3GPP TS 22.186 "Enhancement of 3GPP support for V2X scenarios," v16.1, Dec. 2018.

[4] 3GPP TS 23.285 "Architecture enhancements for V2X services," v15.0.0, Mar. 2018.

[5] 3GPP TR 23.786 "Study on architecture enhancements for EPS and 5G System to support advanced V2X services", v0.6.0, June 2018.

[6] V Jain, S Lapoehn, T Frankiewicz, T Hesse, M Gharba, "Prediction based framework for vehicle platooning using vehicular communications," IEEE Vehicular Networking Conference (VNC), 2017

[7] 3GPP TS 23.287, "Architecture enhancements for 5G System (5GS) to support Vehicle-to-Everything (V2X) services", v1.0.0, June 2019

[8] T.Soni, A.R.Ali, K.Ganesan, M.Schelmann, "Adaptive numerology — A solution to address the demanding QoS in 5G-V2X,", IEEE Wireless Communications and Networking Conference (WCNC). April 2018

[9] K Ganesan, T Soni, S Nunna, AR Ali, "Poster: A TDM approach for latency reduction of ultra-reliable low-latency data in 5G," IEEE Vehicular Networking Conference (VNC), 2016

[10] 3GPP TR 33.836 "Study on security aspects of 3GPP support for advanced V2X service", v0.1.0 May 2019

[11] 3GPP TS 23.303, "Proximity-based services (ProSe); Stage 2, " V15.1, June 2018